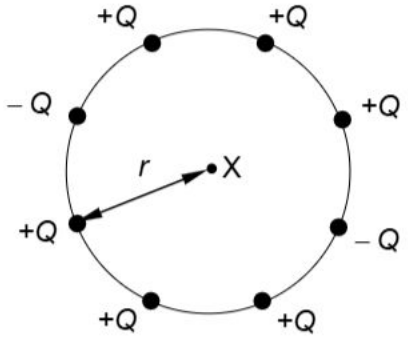
19 A point X is surrounded symmetrically by six positive charges +*Q* and two negative charges −*Q* at distance *r* as shown.



How much work is done by the electric field to bring a point charge of +2*Q* from infinity to position X?

A $-\frac{Q^2}{\pi\varepsilon_0 r}$ **B** $-\frac{2Q^2}{\pi\varepsilon_0 r}$ **C** $\frac{2Q^2}{\pi\varepsilon_0 r}$ **D** $-\frac{4Q^2}{\pi\varepsilon_0 r}$

Ans: **B**

Potential is a scalar.

$$V_{centre} = 6\left(\frac{Q}{4\pi\varepsilon_0 r}\right) + 2\left(\frac{-Q}{4\pi\varepsilon_0 r}\right) = \frac{Q}{\pi\varepsilon_0 r}$$

Work done by external force: $U = (+2Q)(V) = (2Q)\left(\frac{Q}{\pi\varepsilon_0 r}\right) = \frac{2Q^2}{\pi\varepsilon_0 r}$

Hence work done by electric repulsion = $-\frac{2Q^2}{\pi\varepsilon_0 r}$

A: forgot 2Q

C: used WD by external force

D: didn't consider sign of charges